

Assembly & Machine Language Lab

Assignment 1

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1. Write an Assembly Language Program to add two sixteen-bit numbers. The numbers are stored in DS: 0030H and DS: 0040H. Store the result in DS: 0050H, DS: 0051H, and DS: 0052H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov ax,[si]
mov si,0040h
add ax,[si]
mov si,0050h
mov [si],ax
mov ah,0
adc ah,ah
add si,0002h
mov [si],ah
int 03h
mov ah,4ch
int 21h
main endp
end main
```

2. Write an Assembly Language Program to subtract an 8-bit number stored in DS: 0030H from a number stored in DS: 0040H using 2's complement method. Store the result in DS: 0050H, and DS: 0051H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov al,[si]
neg al
```

```

mov si,0040h
add al,[si]
jc l1
neg al
mov si,0050h
mov [si],al
jmp l2
l1:
    mov si,0050h
    mov [si],al
    cmc
    mov ah,00
    adc ah,ah
    inc si
    mov [si],ah
l2:
    int 3
    mov ah,4ch
    int 21h
main endp
end main

```

3. Write a program to transfer a block of 8 data bytes from memory location DS: 0030H to DS: 0040H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov es,ax
mov si,0030h
mov di,0040h
mov cx,0008h
cld
rep movsb
int 03h
mov ah,4ch
int 21h
main endp
end main

```

4. Write an 8086 Assembly Language Program for the addition of 7 eight-bit numbers stored from DS: 0030H. Store the result in DS: 0050H and DS: 0051H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h

```

```

mov di,0050h
mov cx,0007h
mov ax,0000h
mov bx,0000h
l1:  mov bl,[si]
      add ax,bx
      inc si
      loop l1
mov [di], ax
int 03h
mov ah,4ch
int 21h
main endp
end main

```

5. Write an 8086 Assembly Language Program for the addition of 5 sixteen-bit numbers stored from DS: 0030H. Store the result in DS: 0050H, DS: 0051H, DS: 0052H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0050h
mov cx,0005h
mov ax,0000h
mov bx,0000h
mov dl,00h
l1:  mov bx,[si]
      add ax,bx
      adc dl,00h
      inc si
      inc si
      loop l1
mov [di], ax
inc di
inc di
mov [di],dl
int 03h
mov ah,4ch
int 21h
main endp
end main

```

6. Write an Assembly Language Program for the addition of five BCD numbers stored from DS: 0030H. Store the result in DS: 0040H and DS: 0041H.

```

dosseg
.model small
.stack 100h
.data
.code

```

```

main proc
mov ax,@data
mov ds,ax
mov ax,0000h
mov bx,0000h
mov si,0030h
mov di,0040h
mov cx,0005h
l1:
    add al,[si]
    daa
    jnc l2
    inc bl
    l2:
        inc si
    loop l1
mov [di],al
inc di
mov [di],bl
int 03h
mov ah,4ch
int 21h
main endp
end main

```

7. Write an Assembly Language Program to subtract a BCD number stored in DS: 0040H from a BCD number stored in DS: 0050H. Store the result in DS: 0060H and DS: 0061H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov ax,0000h
mov bx,0000h
mov si,0040h
add bl,[si]
mov si,0050h
add al,[si]
mov si,0060h
sub al,bl
das
jnc l1
mov bl,al
mov al,99h
sub al,bl
add al,01h
daa
l1:  mov [si],al
    cmc
    adc ah,ah
    inc si
    mov [si],ah

```

```

int 03h
mov ah,4ch
int 21h
main endp
end main

```

8. Write an Assembly Language Program to multiply two eight bit number stored in DS: 0040H and DS: 0050H. Store the result from DS: 0060H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov ah,00h
mov si,0040h
mov al,[si]
mov si,0050h
mov bl,[si]
mul bl
mov si,0060h
mov [si],ax
int 03h
mov ah,4ch
int 21h
main endp
end main

```

9. Write an Assembly Language Program to multiply two sixteen bit number stored in DS:0040H and DS:0050H. Store the result from DS: 0060H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0040h
mov ax,[si]
mov si,0050h
mov bx,[si]
mul bx
mov si,0060h
mov [si],dx
add si,0002h
mov [si],ax
int 03h
mov ah,4ch
int 21h
main endp
end main

```

10. Write an Assembly Language Program to divide 88H by 33H. Store the quotient in DS: 0060H and remainder in DS: 0061H.

```
dosseg
.model small
.stack 100h
.data
num1 db 88h
num2 db 33h
.code
main proc
mov ax, @data
mov ds, ax
mov al, num1
mov bl, num2
mov ah, 00h
div bl
mov si, 0060h
mov [si], al
mov si, 0061h
mov [si], ah
int 03h
mov ah, 4ch
int 21h
main endp
end main
```

11. Write an Assembly Language Program to divide 2222H by 55H. Store the quotient from DS: 0060H and remainder in DS: 0062H.

```
dosseg
.model small
.stack 100h
.data
dividend dw 2222h
divisor db 55h
.code
main proc
mov ax, @data
mov ds, ax
mov ax, dividend
mov dx, 0000h
mov bl, divisor
div bl
mov si, 0060h
mov [si], ax
mov si, 0062h
mov [si], dx
int 03h
mov ah, 4ch
int 21h
main endp
end main
```